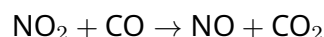


Guided Inquiry: Kinetics in a Plug Flow Reactor

Objective

In this activity, you will use the **PFR Kinetics Simulator** to explore how reaction rate expressions, stoichiometry, and inlet concentrations affect the performance of a Plug Flow Reactor.

Chemical Reaction:



Part 1: Elementary Reactions

Goal: Understand what makes a reaction "elementary" and how it relates to the rate law.

1. Set the conditions:

- Set **Order w.r.t NO₂** to **1.0**.
- Set **Order w.r.t CO** to **1.0**.
- These exponents match the stoichiometric coefficients in the chemical equation (1NO₂+1CO).

2. Observe:

- Look at the red line (NO₂) and the blue line (CO).
- How does the rate of disappearance (the slope of the line) change as you move from left to right (increasing volume)?
- *Question:* Why does the slope get flatter as volume increases?

3. **Definition:** When the exponents in the rate law match the stoichiometric coefficients exactly, the reaction is often called **elementary**. This implies the reaction happens in a single collision step exactly as written.

Part 2: The Chameleon Constant (*k*)

Goal: Discover why the units of the rate constant *k* must change when the rate law changes.

1. **Prediction:** The rate of reaction, *r*, always has units of **concentration / time** (e.g., mol/L/s).

2. Experiment:

- Change the exponents using the sliders and observe the "Unit Display" under the formula box.
- Fill in the table below:

Order (NO ₂)	Order (CO)	Total Order (<i>n</i>)	Units of <i>k</i>
0	0	0	
1	0	1	
1	1	2	
1.5	0.5	2	

3. Analysis:

- Compare the units for the two cases where the Total Order is 2. Are they the same or different?
- *Challenge:* Write a general formula for the units of k in terms of concentration (M), time (s), and total order (n).

Part 3: Stoichiometric Consistency

Goal: Verify that changing the *speed* of the reaction does not change the *ratio* in which chemicals react.

1. Set the conditions:

- Set Inlet $[\text{NO}_2]$ to **1.0 mol/L**.
- Set Inlet $[\text{CO}]$ to **2.0 mol/L**.
- Set Exponents to any value you like (try high values like 2.0, then low values like 0.5).

2. Observe the gap:

- Look at the vertical distance between the Blue line (CO) and the Red line (NO_2).
- Does this vertical distance change as you move down the reactor volume?
- Does this vertical distance change when you make the reaction faster or slower using the exponent sliders?

3. Conclusion:

- The reaction consumes 1 mole of NO_2 for every 1 mole of CO.
- Therefore, the difference $[\text{CO}] - [\text{NO}_2]$ should always be equal to the difference in their initial concentrations: $2.0 - 1.0 = 1.0$.
- *Principle:* Kinetics determines **how fast** we get there; Stoichiometry determines **where** we go.

Part 4: The Power of Concentration

Goal: Visualize how initial concentration drives the reaction rate.

1. Set the conditions:

- Set **Order w.r.t NO_2** to **2.0**.
- Set **Order w.r.t CO** to **0.0**.
- (This effectively makes the rate law: $r = k[\text{NO}_2]^2$).

2. Experiment:

- Start with Inlet $[\text{NO}_2] = 1.0$. Look at the slope of the red line at the very beginning (Volume = 0). It should be moderately steep.
- Increase Inlet $[\text{NO}_2]$ to **2.0** (double the concentration).

3. Analyze the Rate:

- Look at the new slope at Volume = 0. It is much steeper.

- Mathematically, if $r = k[\text{NO}_2]^2$, and you double the concentration ($2x$), by what factor does the initial rate increase?
 - a) $2x$
 - b) $4x$
 - c) $8x$
- Verify this visually by observing how quickly the reactant crashes towards zero compared to the first case.